**Date:** October 28, 2026

**To:** Prof. Mohamed A. Abdou
Editor-in-Chief, *Fusion Engineering and Design* (Elsevier)

**Subject:** Manuscript Submission — *"Shercliff-Hartmann Magnetohydrodynamic Velocity Profiles and Pressure Drop Characteristics in Liquid Metal Divertor Channels"* (Paper IV)

Dear Prof. Mohamed A. Abdou and Editorial Board Members,

I am pleased to submit our original research manuscript entitled **"Shercliff-Hartmann Magnetohydrodynamic Velocity Profiles and Pressure Drop Characteristics in Liquid Metal Divertor Channels"** for publication as a Regular Research Article in *Fusion Engineering and Design.*

## 1. Explicit Selection of Publishing Model

I explicitly select the **Traditional No-Fee Subscription Publishing Model** for this submission. Under this publishing pathway, no Article Processing Charges (APCs) or publication fees apply.

## 2. Scientific Scope & Core Novelty

This work formulates a self-contained 3D liquid metal magnetohydrodynamic (MHD) boundary layer framework for eutectic lead-lithium (Pb-17Li) flows in high-field divertor channels ($B_0 = 10$ T, $Ha = 10 \ldots 1000$). Key contributions include:

- **Boundary Layer Thinning:** High-resolution numerical solution demonstrating Hartmann boundary layer compression down to $\delta_H = a/Ha = 0.20$ mm at $Ha = 500$.
- **MHD Pressure Drop Modeling:** Quantification of pressure drop gradients ($dP/dz = 0.45$ MPa/m) and velocity core flattening.
- **Open-Science Reproducibility:** Full deployment of Python solvers, raw sweep matrices, and automated CI/CD validation scripts.

## 3. Declarations

- **Originality:** This manuscript represents original, self-contained work that has not been published previously and is not under consideration elsewhere.
- **Conflicts of Interest:** The author declares no competing financial or personal interests.
- **AI Disclosure:** Scientific theory, mathematical proofs, and core physics were authored directly by the investigator. Generative AI tools were used solely for code syntax optimization in data plotting scripts.

Thank you for considering this manuscript for publication in *Fusion Engineering and Design.*

Sincerely,

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